THAKUR COLLEGE OF ENGINEERING & TECHNOLOGY
(Approved by AICTE, Govt. of Maharashtra & Affiliated to University of Mumbai)
A - Block, Thakur Educational Campus, Shyamnayaran Thakur Marg,
Thakur Village, Kandivali (East), Mumbai - 400 101.

IN-SEMESTER EXAMINATION-II

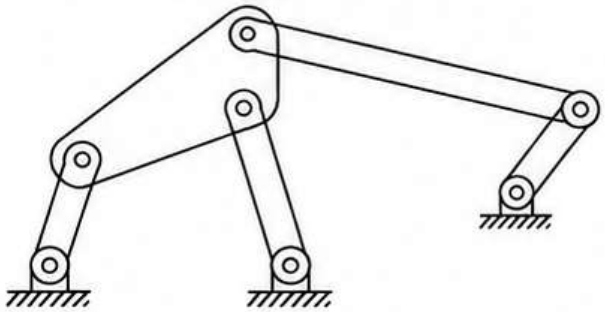
F.E./F.T. (Semester-I)

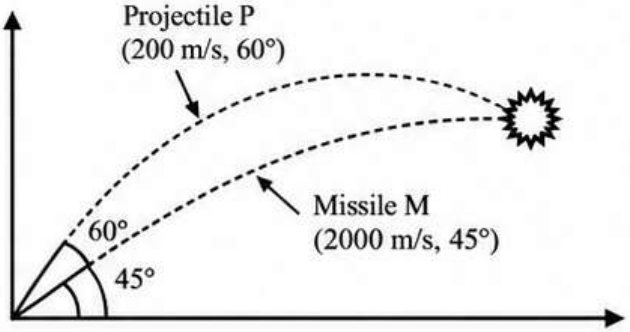
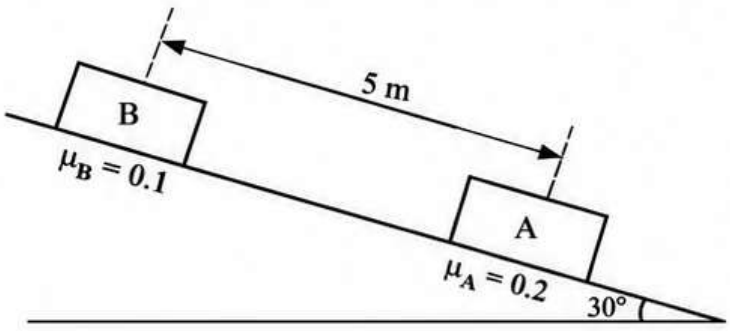
SUBJECT –Engineering Mechanics

Branch: AIML, E&TC, E&CS, IT, MECH, MME **Date: 23/12/2025**
Div.: ALL **Timing: 4:00 PM to 5:00 PM**
Duration: 60 Minutes **Maximum Marks: 20**

Instructions –

1. All questions are compulsory.
2. Assume suitable data wherever necessary and state the assumptions made.
3. Diagrams / sketches should be given wherever necessary.
4. Use of logarithmic table, drawing instruments and non-programmable calculators is permitted.
5. Figures to the right indicate full marks.

Q.1		Answer any 5 of the following questions	Marks	Course Outcomes	Learning Levels
	a.	Define Manipulator in robotics.	2	CO4	R
	b.	Define General Plane Motion.	2	CO5	R
	c.	Find the DOF for the following mechanism. 	2	CO4	A
	d.	Write Formulae to determine Range, Time of Flight.	2	CO5	R
	e.	A particle freely falling under gravity falls 30m in a certain second. Find the velocity at a height at 30m.	2	CO5	A
	f.	Enlist Kinematical equation and provide condition to apply.	2	CO6	R
	g.	State D' Alembert's Principle.	2	CO6	R
Q.2	a.	Frame A is translated by (3,0,0) then rotated 90° about Z. Frame B is attached to A and has point (1,0,0). What is the global position?	5	CO4	An
		OR			
	b.	Furniture is seen at (2, −1) in robot's local frame. Robot is at (6,4) facing 270°. Where is the furniture in the room?	5	CO4	An

Q.3	a.	A projectile P is fired at a muzzle velocity of 200 m/s at an angle of elevation of 60°, as shown in Fig. After some time, a missile M is fired at 2000 m/s muzzle velocity and at an angle of elevation of 45° from the same point to destroy the projectile P. Find the horizontal distance.  	5	CO5	A
		OR			
	b.	Two blocks A and B are held on an inclined plane 5 m apart as shown in figure. Take $\mu_A = 0.2$ and $\mu_B = 0.1$. If the blocks begin to slide down the plane simultaneously calculate the time and distance traveled by each block before collision if the acceleration of the block A & B is 1.512 m/s^2 & 2.43 m/s^2 respectively along the slope.  	5	CO6	A